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Computers and Electronics in Agriculture

Foundation Models in Agriculture: Advancing AI for Precision and Smart Farming

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Dear Editors,

We are pleased to submit our manuscript entitled “**Quantifying the contribution of multimodal geospatial context in foundation models for parcel-level agricultural decision support**” for consideration for publication in the special issue *Foundation Models in Agriculture: Advancing AI for Precision and Smart Farming* in *Computers and Electronics in Agriculture*.

This manuscript is a substantially extended version of our conference paper presented at the *2025 4th International Conference on Geographic Information and Remote Sensing Technology (GIRST)* (DOI: 10.1109/GIRST67753.2025.11382141) which proposed a multimodal LLM-based system with geospatial augmented context for agricultural plot-level assistance. The journal version expands the conference contribution by shifting the focus from the system’s technical implementation to a comprehensive, mathematically grounded evaluation of its outputs, providing both a reproducible methodology for quantifying architectural modules and the dataset built for the study.

This paper addresses a critical gap in the systematic evaluation of multimodal language models augmented with Earth observation multi-domain data for precision agriculture and land management. Existing evaluation frameworks fail to measure how these models actually utilize dynamically retrieved context. To solve this, we propose a novel, reproducible evaluation methodology that quantifies the marginal contributions of distinct architectural modules by applying collaborative game theory analysis. We showcase this framework by applying it to the aforementioned validated context-amplified LLM, yielding very significant findings on both the system architecture and the methodological framework itself.

Our approach introduces a hybrid human-in-the-loop panel combining two human evaluators and two distinct LLM judges, to assess a 143-query dataset across five targeted metrics. By designing a full factorial ablation study modeled as a cooperative game and applying Shapley values, we quantify the impact of Satellite, Orthophoto, and Retrieval-Augmented Generation (RAG) modules. We show that visual data drive up to a 32% increase in overall response utility for given queries, while RAG module exhibits high, query-dependent volatility. Furthermore, our context length analysis proves the architecture’s resilience to information overload, while inter-evaluator consensus metrics highlight the continued necessity of human oversight to adequately penalize bias.

To ensure reproducibility and open science, the complete evaluation dataset, containing agricultural parcel orthophotos, retrieved external context, and generated responses, is shared supplementary material.

We believe this study aligns perfectly with the scope of *Computer and Electronics in Agriculture*, and would be a particularly strong fit for the special issue *Foundation Models in Agriculture: Advancing AI for Precision and Smart Farming*. We feel that our findings will be highly relevant to your audience, providing a solid contribution to the research community through the establishment of a rigorous, mathematical standard for benchmarking next-generation foundation models in smart farming.

We confirm that this manuscript is original, has not been published elsewhere, and is not currently under consideration by any other journal. All authors have approved the manuscript and agree with its submission. The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

We thank you for considering our work and look forward to your editorial decision.

Sincerely,

Prof. J. Molleda, on behalf of all authors.